Temperature Converter

Programming Assignment — README

Assignment	Temperature Converter
Course	Java Programming
Due Date	June 7, 2026 at 11:59 PM
Submission	Push your .java file to the course GitHub repository
File Required	TemperatureConverter.java (or similar)

Overview

Write a Java console program that repeatedly prompts the user for a temperature and its unit (Celsius or Fahrenheit), converts it to the opposite unit, and displays the result. The program continues looping until the user types "stop".

Requirements

Core Behavior

- Prompt the user to enter a temperature value or type "stop" to quit.
- Prompt the user for the unit (C or F).
- Convert and display the result in the format shown in the examples below.
- Loop until the user types "stop".

Error Handling

- Display a meaningful error message for invalid temperature inputs (e.g., letters instead of numbers).
- Display a meaningful error message for unrecognized unit labels.
- Re-prompt the user after any invalid input — do not crash or exit.
- Do not use **exceptions, break statements, or try/catch blocks**.

Required Method

Your program must include the following method signature exactly:

```
public static double convertTemperature(double temperature, String unit)
```

- Receives the temperature value and the unit type as arguments.

- Returns the converted temperature as a double.

Conversion Formulas

Use these formulas inside your convertTemperature method:

Conversion	Formula
Celsius → Fahrenheit	`F = (C × 9/5) + 32`
Fahrenheit → Celsius	`C = (F − 32) × 5/9`

Expected Output Examples

Input	Expected Console Output
`100` / `C`	`100.00°C is equal to 212.00°F`
`32` / `F`	`32.00°F is equal to 0.00°C`

Test Cases & Grading

⚠️ **IMPORTANT: All test cases must pass with a green check mark** in order to receive full credit for this assignment. **Submissions that do not produce a green check mark will not receive credit.**

Your program will be run against the following test cases. Each test must produce the correct converted output and handle the corresponding input correctly:

#	Temp Input	Unit	Expected Output	Pass Condition
1	`100`	`C`	`100.00°C is equal to 212.00°F`	✅ Green check
2	`32`	`F`	`32.00°F is equal to 0.00°C`	✅ Green check
3	`0`	`C`	`0.00°C is equal to 32.00°F`	✅ Green check
4	`-40`	`C`	`-40.00°C is equal to -40.00°F`	✅ Green check
5	`abc`	—	`Error message displayed, re-prompt shown`	✅ Green check
6	`100`	`X`	`Error message displayed, re-prompt shown`	✅ Green check
7	`stop`	—	`Program exits gracefully`	✅ Green check

Constraints & Restrictions

- **No exceptions:** Do not use try/catch, throw, or the Exception class.
- **No break statements:** Control all loops using boolean conditions only.
- **Input validation:** The program must not crash under any input. Test it thoroughly with letters, symbols, and blank entries.
- **Method signature:** The convertTemperature method must match the required signature exactly.

Submission Instructions

- Push your `.java` file to the course GitHub repository.
- Make sure your commit includes your final, working code — not a work in progress.
- Ensure your file compiles and runs without errors before pushing.
- All automated test cases must show a **green check mark** ✅ to receive credit.
- Due: **June 7, 2026 at 11:59 PM**
